

Making up infant formula at a Target concentration

Name _____ Date _____

Use a basic calculator and the cheat sheet. Show your working. Round as the cheat sheet says: scoops up to a whole scoop, water up to the nearest 5 mL.

1 Reading a label

Work out the displacement and kcal per scoop of Formula A.

Formula A: the label says

Energy: 68 kcal per 100 mL

Reconstitution statement: 2 scoops + 60 mL water makes approximately 67 mL

Grams per scoop: 1 scoop = 4.6 g

Step	Working	Answer
A1 Scoops in the statement		scoops
A2 Displacement (mL per g)		mL per g
A3 kcal per scoop		kcal

2 Reading a label

Work out the displacement and kcal per scoop of Formula B.

Formula B: the label says

Energy: 66 kcal per 100 mL

Reconstitution statement: 132 g powder + 900 mL water makes 1 L

Grams per scoop: Average scoop weight = 4.4 g

Step	Working	Answer
A1 Scoops in the statement		scoops
A2 Displacement (mL per g)		mL per g
A3 kcal per scoop		kcal

3 Building a recipe

Formula C has **21.5 kcal per scoop**, **4.5 g per scoop** and a displacement of **0.711 mL per g**. Make **180 mL** at **20 kcal/30 mL**.

Step	Working	Answer
1	kcal/mL	
2	kcal needed	kcal
3	Scoops	scoops
4	Actual kcal	kcal
5	Total powder (g)	g
6	Total displacement (mL)	mL
7	Water (mL)	mL
8	Actual volume (mL)	mL
9	Actual concentration (kcal/30 mL)	kcal/30 mL

Recipe: _____ scoops + _____ mL water

Makes: _____ mL at _____ kcal/30 mL

0.5 kcal/30 mL or more under target? Yes / No

4 Building a recipe

Formula C has **21.5 kcal per scoop**, **4.5 g per scoop** and a displacement of **0.711 mL per g**. Make **150 mL** at **27 kcal/30 mL**.

	Step	Working	Answer
1	kcal/mL		
2	kcal needed		kcal
3	Scoops		scoops
4	Actual kcal		kcal
5	Total powder (g)		g
6	Total displacement (mL)		mL
7	Water (mL)		mL
8	Actual volume (mL)		mL
9	Actual concentration (kcal/30 mL)		kcal/30 mL

Recipe: _____ scoops + _____ mL water

Makes: _____ mL at _____ kcal/30 mL

0.5 kcal/30 mL or more under target? Yes / No

5 Building a recipe

Formula D has **36 kcal per scoop, 7 g per scoop** and a displacement of **0.714 mL per g**. Make **60 mL at 30 kcal/30 mL**.

	Step	Working	Answer
1	kcal/mL		
2	kcal needed		kcal
3	Scoops		scoops
4	Actual kcal		kcal
5	Total powder (g)		g
6	Total displacement (mL)		mL
7	Water (mL)		mL
8	Actual volume (mL)		mL
9	Actual concentration (kcal/30 mL)		kcal/30 mL

Recipe: _____ scoops + _____ mL water

Makes: _____ mL at _____ kcal/30 mL

0.5 kcal/30 mL or more under target? Yes / No

6 From label to bottle

Make 180 mL at 24 kcal/30 mL of Formula E.

Formula E: the label says

Energy: 67 kcal per 100 mL

Reconstitution statement: 135 g powder + 900 mL water makes 1 L

Grams per scoop: Average scoop weight = 4.5 g

Part A • Reading the label

Step	Working	Answer
A1	Scoops in the statement	scoops
A2	Displacement (mL per g)	mL per g
A3	kcal per scoop	kcal

Part B • Building the recipe

Step	Working	Answer
1	kcal/mL	
2	kcal needed	kcal
3	Scoops	scoops
4	Actual kcal	kcal
5	Total powder (g)	g
6	Total displacement (mL)	mL
7	Water (mL)	mL
8	Actual volume (mL)	mL
9	Actual concentration (kcal/30 mL)	kcal/30 mL

Recipe: _____ scoops + _____ mL water

Makes: _____ mL at _____ kcal/30 mL

0.5 kcal/30 mL or more under target? Yes / No

7 From label to bottle

Make 120 mL at 27 kcal/30 mL of Formula F.

Formula F: the label says

Energy: 70 kcal per 100 mL

Reconstitution statement: 1 scoop + 30 mL water makes approximately 33 mL

Grams per scoop: 1 scoop = 4.4 g

Part A • Reading the label

Step	Working	Answer
A1	Scoops in the statement	scoop
A2	Displacement (mL per g)	mL per g
A3	kcal per scoop	kcal

Part B • Building the recipe

Step	Working	Answer
1	kcal/mL	
2	kcal needed	kcal
3	Scoops	scoops
4	Actual kcal	kcal
5	Total powder (g)	g
6	Total displacement (mL)	mL
7	Water (mL)	mL
8	Actual volume (mL)	mL
9	Actual concentration (kcal/30 mL)	kcal/30 mL

Recipe: _____ scoops + _____ mL water

Makes: _____ mL at _____ kcal/30 mL

0.5 kcal/30 mL or more under target? Yes / No

8 Explain

1. Why isn't the water simply the Target volume?

2. Why are scoops rounded up rather than to the nearest whole scoop?

3. A recipe's Actual concentration comes out 0.5 kcal/30 mL or more below the Target concentration. What should you do?
